

#03 Smart Home Energy Saver Agent

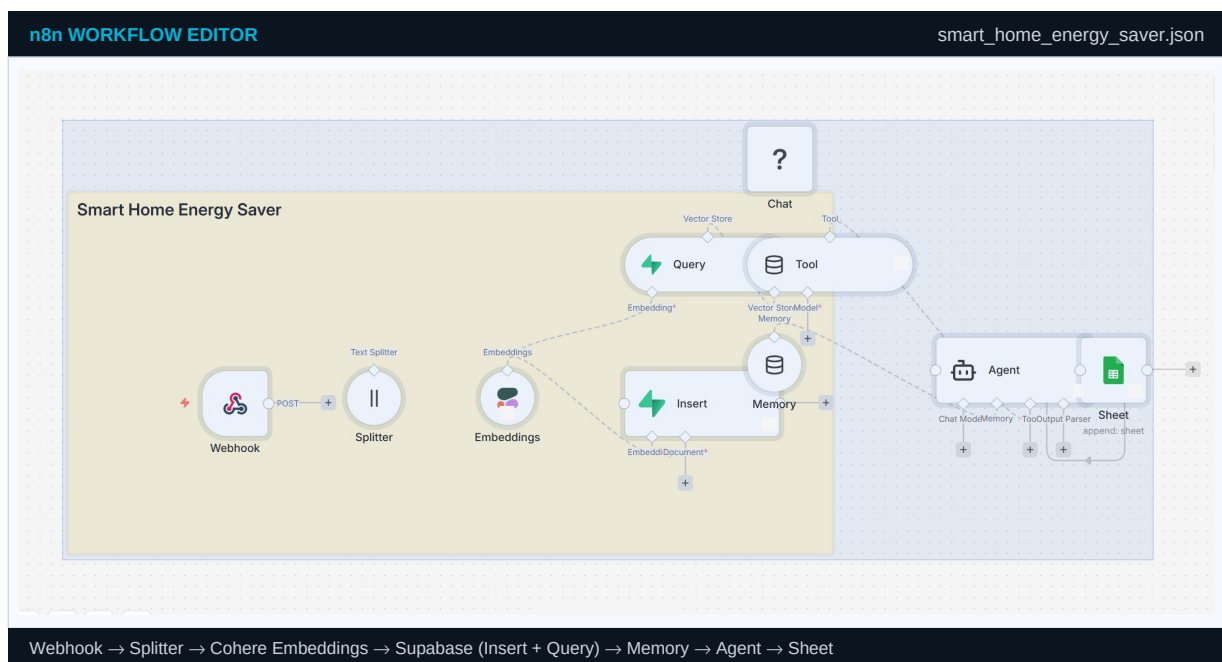
IoT / Smart Energy

AI learns your consumption patterns and cuts waste autonomously

BUSINESS PROBLEM

Smart plugs generate enormous data but most homeowners never act on it. This AI agent autonomously analyses consumption patterns, detects waste such as AC running with no occupancy or standby loads overnight, controls devices via HTTP calls to Home Assistant, and delivers weekly savings reports.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — smart plug reading (Home Assistant, every 5 min)

```
{
  "device_id": "socket_living_ac",
  "room": "Living Room",
  "watts": 1840,
  "voltage": 230,
  "occupancy": false,
  "timestamp": "2024-06-15T00:35:00Z",
  "tariff_band": "off-peak"
}
```

OUTPUT — Agent output — device control action + weekly log entry

```
{
  "json": {
    "device_id": "socket_living_ac",
    "action": "turn_off",
    "reason": "No occupancy detected for 48 minutes",
    "kwh_saved": 0.147,
    "cost_saved": 0.022,
    "ha_command": {
      "entity_id": "switch.living_ac",
      "service": "switch.turn_off"
    },
    "weekly_total_kwh_saved": 14.2
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — 5-min smart plug reading	Home Assistant fires: {device_id, room, watts, voltage, occupancy_sensor, timestamp, tariff_band}.
2	➤ Splitter — energy guidelines chunks	UK/regional energy efficiency guides and device spec sheets split for RAG. 'Daikin FTXS35 rated 1200W cooling, idle standby 8W.'
3	■ Cohere Embeddings — device profiles	Device power profiles embedded: rated wattage, normal operating range, standby draw for each device type on the system.
4	■ Supabase Insert — consumption history	Every reading stored. Agent queries: AVG kWh by hour-of-day per device to build behavioural baseline.
5	■ Supabase Query — pattern retrieval	'Show times AC ran more than 2h with no occupancy detected' — identifies recurring waste windows.
6	■ Vector Tool — efficiency standards	Agent checks: is this device drawing more than rated spec? If yes, flag for inspection alongside waste detection.
7	■ Window Memory — 7-day baseline	Rolling weekly baseline. Today vs weekly average → anomaly detection for unusual consumption events.
8	■ HuggingFace LLM — decision making	Evaluates: AC on, no motion 45min → turn off. Standby loads more than 5 devices → cut at midnight schedule.
9	■ Agent — executes device control	HTTP POST to Home Assistant: {entity_id, service: switch.turn_off / climate.set_temperature}. Fully autonomous.
10	■ Sheets — weekly savings tracker	Logs kWh and cost saved per action. Weekly auto-report: total_savings_kWh, cost_saved, top_actions list.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Cohere API Key	dashboard.cohere.com → API Keys	Cohere API
Supabase URL + Anon Key	app.supabase.com → Settings → API	Supabase API
Home Assistant Long-Lived Token	HA → Profile → Long-Lived Access Tokens	HTTP Header Auth
Google Sheets OAuth2	console.cloud.google.com	Google Sheets OAuth2

SAMPLE RUN — what this workflow processed and produced

Scenario	A 3-bedroom house in a hot climate runs a split AC unit in the living room. The homeowner leaves for work at 07:30 but often forgets to turn off the AC. Electricity cost is \$0.15/kWh.
Input received	At 00:35, socket_living_ac posted 1840W draw. Occupancy sensor: false. AC had been running since 22:47 — 108 minutes with no occupancy detected.
What the workflow found	Supabase baseline showed this device averages 1,820W when running. No occupancy for 48+ minutes exceeded the configured waste threshold. Weekly pattern showed this happens 4-5 nights per week.

Output produced	Agent sent turn_off command to Home Assistant at 00:35. Estimated saving for this single event: 0.147 kWh (\$0.022). Weekly total saved in the same run: 14.2 kWh (\$2.13). Monthly projection: 61.5 kWh (\$9.23).	
JSON: smart_home_energy_saver.json	GitHub: trm-smart-energy-agent	www.home-assistant.io/integrations/